WEB DEVELOPMENT PROJECT REPORT

NRI UNIVERSITY PORTAL (ACADEMIC YEAR 2026-27)

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1. INTRODUCTION

Web development plays a pivotal role in establishing modern digital interfaces for institutions, businesses, and educational entities. An official university web portal acts as the digital storefront and primary bridge between prospective students, enrolled candidates, faculty members, and administrative bodies.

This project focuses on the end-to-end design and responsive front-end implementation of the NRI University web application. Built using clean, semantic HyperText Markup Language (HTML5) and modular Cascading Style Sheets (CSS3), the portal delivers an intuitive user experience. It provides vital institutional resources across dedicated web pages: the Home Page (index.html), the Academic Offerings Directory (courses.html), and the Admissions & Student Onboarding Gateway (admissions.html).

2. AIM

To design, develop, and implement a multi-page, responsive, and accessible static university website for NRI University using fundamental front-end web technologies (HTML5 and CSS3) that delivers clear institutional information, showcases academic courses, outlines admission guidelines, and provides an interactive admissions enquiry channel.

3. OBJECTIVES

- **Institutional Identity & Branding:** Establish a cohesive visual theme matching institutional values using structured color schemes (Navy #123b70, Deep Slate #08264b, and Warm Gold #ffb703) and clean typography.
- **Multi-Page Architecture:** Construct seamless navigation across Home, Courses, and Admissions pages using interconnected relative anchor hyperlinks.
- **Card-Based Information Architecture:** Utilize CSS Flexbox to present academic departments, degree programs, faculty profiles, campus facilities, and upcoming events in clean, digestible cards.
- **User Interactive Channels:** Provide a comprehensive Admission Enquiry form with structured validation fields (Student Name, Email Address, Program Selection, and Message).
- **Cross-Device Responsiveness:** Incorporate CSS Media Queries to adapt visual layouts seamlessly between desktop monitors, tablets, and mobile screens without breaking content hierarchy.

4. HTML CONCEPTS USED

HTML5 provides the foundational structure and semantics of the web application. The core elements and concepts deployed include:

HTML Element / Tag	Implementation Role in NRI University Portal

`<!DOCTYPE html> & <html>`	Declares modern HTML5 standard compliance and root document structure.
`<header> & <nav>`	Semantic landmarks defining university brand headers and global navigation menus.
`<section> & .container`	Logical content division encapsulating hero banners, departments, and procedures.
`<div> & .cards / .card`	Component-level structural containers powering modular card layouts.
`<img> Tag`	Embedding responsive visual assets with relevant alternate text attributes (alt).
`<form>, <input>, <select>, <textarea>`	Interactive form controls for prospective student enquiries with required field validations.
`<ul class='facilities'> & <li>`	Unordered list elements organizing campus facility overviews.
`<footer>`	Standard bottom semantic landmark containing copyright and institutional ownership data.

5. CSS CONCEPTS USED

Cascading Style Sheets (CSS3) govern the visual aesthetics, layout distribution, and responsiveness of the NRI University portal:

CSS Rule / Property	Functional Purpose & Styling Behavior
Universal Box Model Reset	* { margin: 0; padding: 0; box-sizing: border-box; } ensures precise element dimensional control without margin discrepancies.
Flexbox Layout Model	.cards { display: flex; flex-wrap: wrap; justify-content: center; gap: 20px; } provides automatic multi-column alignment and fluid wrapping.
Hero Banner & Linear Gradient	.banner combines a dual-stop RGBA overlay linear-gradient(rgba(0,30,70,0.65), rgba(0,30,70,0.65)) with background-size: cover for high legibility.
Card Elevation & Aesthetics	.card incorporates border-radius: 8px, overflow: hidden, and box-shadow: 0 3px 10px #cccccc to achieve

	modern material elevation.
Media Queries (@media)	@media (max-width: 600px) dynamically scales typography, restacks navigation links, and expands card widths to 100% for smartphone displays.
Interactive Pseudo-Classes	nav a:hover { color: #ffd166; } and form button:hover { background-color: #2165a8; } offer visual feedback on user cursor interactions.

6. PROCEDURE / DEVELOPMENT STEPS

Step 1: Requirement Gathering & Content Architecture Planning: Identified key institutional facets required for NRI University, establishing a structured 3-page site map: Home (institutional overview, departments, campus life), Courses (academics, faculty, labs/facilities), and Admissions (4-step admission pipeline, upcoming events, contact information, enquiry form).

Step 2: Workspace Setup & Directory Structure: Configured a clean, unified project directory holding all source files (index.html, courses.html, admissions.html, style.css) and verified uniform relative resource linking across all documents.

Step 3: Global Styling & Design System Specification (style.css): Drafted the universal CSS stylesheet. Defined root color variables (Primary Navy #123b70, Dark Slate #08264b, Gold Accent #ffb703, Light Slate Canvas #f4f7fb), unified font families (Arial, sans-serif), responsive container constraints (max-width: 1100px), and card grid models.

Step 4: Home Page Implementation (index.html): Built the university landing page with an introductory institutional banner, department showcases (Computer Science, Electronics Engineering, Mechanical Engineering), and campus highlights utilizing curated imagery.

Step 5: Courses & Faculty Module Implementation (courses.html): Structured academic catalog cards showcasing 4-year degree programs (B.Tech CSE, CSE - AI & ML, Electrical Engineering), introduced department faculty profiles, and highlighted on-campus learning facilities.

Step 6: Admissions Gateway & Form Integration (admissions.html): Outlined the sequential 4-step admission procedure cards, integrated upcoming institutional events (Technical Seminar, College Fest, Workshop), provided institutional contact coordinates, and coded the front-end Admission Enquiry Form.

Step 7: Responsive Refinement & Cross-Browser Verification: Implemented CSS media queries targeting mobile breakpoints (max-width: 600px). Tested viewport responsiveness, form validation inputs, anchor navigation paths, and verified seamless layout rendering across modern web browsers.

7. SOURCE CODE

Below is the complete source code for all project files comprising the NRI University web application:

7.1 index.html (Home Page)

```
<!DOCTYPE html>
```

```
<html>
<head>
    <title>NRI University</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

<header>
    <h1>NRI UNIVERSITY</h1>
    <p>Learn Today, Lead Tomorrow</p>
</header>

<nav>
    <a href="index.html">Home</a>
    <a href="courses.html">Courses</a>
    <a href="admissions.html">Admissions</a>
</nav>

<section class="banner">
    <h2>Welcome to NRI University</h2>
    <p>Build Your Future With Quality Education</p>
    <a class="btn" href="admissions.html">Apply Now</a>
</section>

<section class="container">
    <h2>About Our University</h2>
    <p>
        NRI University provides a learning environment where students
        can develop knowledge, technical skills and creativity.
        Our goal is to prepare students for a successful future.
    </p>
</section>

<section class="container">
    <h2>Our Departments</h2>

    <div class="cards">
        <div class="card">
            <img src="https://images.unsplash.com/photo-1518770660439-
4636190af475?auto=format&fit=crop&w=600&q=80" alt="Computer hardware">
            <h3>Computer Science</h3>
            <p>Learn programming, software development and AI.</p>
        </div>

        <div class="card">
            <img src="https://images.unsplash.com/photo-1581092921461-
eab62e97a780?auto=format&fit=crop&w=600&q=80" alt="Engineering equipment">
            <h3>Electronics Engineering</h3>
            <p>Explore electronic circuits and communication systems.</p>
        </div>

        <div class="card">
            <img src="https://images.unsplash.com/photo-1581092160562-
40aa08e78837?auto=format&fit=crop&w=600&q=80" alt="Engineering laboratory">
            <h3>Mechanical Engineering</h3>
```

```
            <p>Study machines, manufacturing and design.</p>
        </div>
    </div>
</section>

<section class="container">
    <h2>Campus Life</h2>

    <div class="cards">
        <div class="card">
            <img src="https://images.unsplash.com/photo-1562774053-
701939374585?auto=format&fit=crop&w=600&q=80" alt="University building">
            <h3>Modern Campus</h3>
            <p>A welcoming environment for learning and growth.</p>
        </div>

        <div class="card">
            <img src="https://images.unsplash.com/photo-1507842217343-
583bb7270b66?auto=format&fit=crop&w=600&q=80" alt="Library">
            <h3>Library</h3>
            <p>A space for reading, studying and gaining knowledge.</p>
        </div>

        <div class="card">
            <img src="https://images.unsplash.com/photo-1523580494863-
6f3031224c94?auto=format&fit=crop&w=600&q=80" alt="Students on campus">
            <h3>Student Activities</h3>
            <p>Participate in events and student activities.</p>
        </div>
    </div>
</section>

<footer>
    <p>2026 NRI University | All Rights Reserved</p>
</footer>

</body>
</html>
```

7.2 courses.html (Courses and Faculty Page)

```
<!DOCTYPE html>
<html>
<head>
    <title>Courses - NRI University</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

<header>
    <h1>NRI UNIVERSITY</h1>
    <p>Courses and Faculty</p>
</header>
```

```
<nav>
    <a href="index.html">Home</a>
    <a href="courses.html">Courses</a>
    <a href="admissions.html">Admissions</a>
</nav>

<section class="container">
    <h2>Our Courses</h2>

    <div class="cards">
        <div class="card">
            <img src="https://images.unsplash.com/photo-1516321318423-
f06f85e504b3?auto=format&fit=crop&w=600&q=80" alt="Computer programming">
            <h3>B.Tech CSE</h3>
            <p>Learn programming, algorithms and computer science.</p>
            <p><b>Duration:</b> 4 Years</p>
        </div>

        <div class="card">
            <img src="https://images.unsplash.com/photo-1485827404703-
89b55fcc595e?auto=format&fit=crop&w=600&q=80" alt="Artificial intelligence robot">
            <h3>CSE - AI & ML</h3>
            <p>Learn artificial intelligence, machine learning and data science.</p>
            <p><b>Duration:</b> 4 Years</p>
        </div>

        <div class="card">
            <img src="https://images.unsplash.com/photo-1581092160607-
ee22621dd758?auto=format&fit=crop&w=600&q=80" alt="Electrical engineering">
            <h3>Electrical Engineering</h3>
            <p>Learn electrical systems, machines and power technology.</p>
            <p><b>Duration:</b> 4 Years</p>
        </div>
    </div>
</section>

<section class="container">
    <h2>Our Faculty</h2>

    <div class="cards">
        <div class="card">
            <img src="https://images.unsplash.com/photo-1580894732444-
8ecded7900cd?auto=format&fit=crop&w=600&q=80" alt="Faculty member">
            <h3>Computer Science Faculty</h3>
            <p>Programming and Software Development</p>
        </div>

        <div class="card">
            <img src="https://images.unsplash.com/photo-1560250097-
0b93528c311a?auto=format&fit=crop&w=600&q=80" alt="Faculty member">
            <h3>Engineering Faculty</h3>
            <p>Engineering and Technical Education</p>
        </div>
```

```
        <div class="card">
            <img src="https://images.unsplash.com/photo-1551836022-
d5d88e9218df?auto=format&fit=crop&w=600&q=80" alt="Faculty member">
            <h3>Research Faculty</h3>
            <p>Research and Innovation</p>
        </div>
    </div>
</section>

<section class="container">
    <h2>Campus Facilities</h2>
    <ul class="facilities">
        <li>Library and reading rooms</li>
        <li>Computer laboratories</li>
        <li>Sports and games</li>
        <li>Seminar halls</li>
        <li>Student activity spaces</li>
    </ul>
</section>

<footer>
    <p>&copy; 2026 NRI University | All Rights Reserved</p>
</footer>

</body>
</html>
```

7.3 admissions.html (Admissions & Enquiry Page)

```
<!DOCTYPE html>
<html>
<head>
    <title>Admissions - NRI University</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>

<header>
    <h1>NRI UNIVERSITY</h1>
    <p>Your Journey Starts Here</p>
</header>

<nav>
    <a href="index.html">Home</a>
    <a href="courses.html">Courses</a>
    <a href="admissions.html">Admissions</a>
</nav>

<section class="banner">
    <h2>Admissions 2026-27</h2>
    <p>Take the First Step Towards Your Dream Career</p>
</section>
```

```
<section class="container">
    <h2>Admission Procedure</h2>

    <div class="cards">
        <div class="card">
            <h3>Step 1</h3>
            <p>Choose your preferred course.</p>
        </div>

        <div class="card">
            <h3>Step 2</h3>
            <p>Check the eligibility requirements.</p>
        </div>

        <div class="card">
            <h3>Step 3</h3>
            <p>Submit the application and required documents.</p>
        </div>

        <div class="card">
            <h3>Step 4</h3>
            <p>Complete the admission process after approval.</p>
        </div>
    </div>
</section>

<section class="container">
    <h2>Upcoming Events</h2>

    <div class="cards">
        <div class="card">
            <img src="https://images.unsplash.com/photo-1540575467063-
178a50c2df87?auto=format&fit=crop&w=600&q=80" alt="Technical seminar">
            <h3>Technical Seminar</h3>
            <p>Explore new technologies and engineering ideas.</p>
        </div>

        <div class="card">
            <img src="https://images.unsplash.com/photo-1517457373958-
b7bdd4587205?auto=format&fit=crop&w=600&q=80" alt="College event">
            <h3>College Fest</h3>
            <p>Enjoy cultural activities and student performances.</p>
        </div>

        <div class="card">
            <img src="https://images.unsplash.com/photo-1523240795612-
9a054b0db644?auto=format&fit=crop&w=600&q=80" alt="Students working together">
            <h3>Student Workshop</h3>
            <p>Develop practical skills through group activities.</p>
        </div>
    </div>
</section>

<section class="container">
    <h2>Contact Us</h2>
```

```
    <div class="contact">
        <p><b>College:</b> NRI University</p>
        <p><b>Address:</b> Add the official university address here</p>
        <p><b>Email:</b> Add the official university email here</p>
        <p><b>Phone:</b> Add the official contact number here</p>
    </div>
</section>

<section class="container">
    <h2>Admission Enquiry</h2>

    <form>
        <label>Student Name</label>
        <input type="text" placeholder="Enter your name" required>

        <label>Email Address</label>
        <input type="email" placeholder="Enter your email" required>

        <label>Select Course</label>
        <select required>
            <option value="">Choose a course</option>
            <option>B.Tech CSE</option>
            <option>CSE - AI & ML</option>
            <option>Electrical Engineering</option>
        </select>

        <label>Message</label>
        <textarea placeholder="Enter your enquiry"></textarea>

        <button type="submit">Submit Enquiry</button>
    </form>
</section>

<footer>
    <p> 2026 NRI University | All Rights Reserved</p>
</footer>

</body>
</html>
```

7.4 style.css (Universal Cascading Stylesheet)

```
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: Arial, sans-serif;
    background-color: #f4f7fb;
    color: #333;
```

```
    line-height: 1.6;
}

header {
    background-color: #123b70;
    color: white;
    text-align: center;
    padding: 25px;
}

header h1 {
    font-size: 32px;
}

nav {
    background-color: #08264b;
    text-align: center;
    padding: 15px;
}

nav a {
    color: white;
    text-decoration: none;
    margin: 15px;
    font-weight: bold;
}

nav a:hover {
    color: #ffd166;
}

.banner {
    background-image: linear-gradient(
        rgba(0, 30, 70, 0.65),
        rgba(0, 30, 70, 0.65)
    ), url("https://images.unsplash.com/photo-1562774053-
701939374585?auto=format&fit=crop&w=1600&q=80");
    background-size: cover;
    background-position: center;
    color: white;
    text-align: center;
    padding: 100px 20px;
}

.banner h2 {
    font-size: 36px;
    margin-bottom: 10px;
}

.btn {
    display: inline-block;
    background-color: #ffb703;
    color: #222;
    padding: 10px 22px;
    margin-top: 20px;
```

```
    text-decoration: none;
    border-radius: 5px;
    font-weight: bold;
}

.container {
    max-width: 1100px;
    margin: auto;
    padding: 35px 20px;
}

.container h2 {
    color: #123b70;
    text-align: center;
    margin-bottom: 20px;
}

.container > p {
    text-align: center;
}

.cards {
    display: flex;
    flex-wrap: wrap;
    justify-content: center;
    gap: 20px;
}

.card {
    background-color: white;
    width: 300px;
    border-radius: 8px;
    overflow: hidden;
    box-shadow: 0 3px 10px #cccccc;
    padding-bottom: 20px;
    text-align: center;
}

.card img {
    width: 100%;
    height: 190px;
    object-fit: cover;
}

.card h3 {
    color: #123b70;
    margin: 15px 10px 8px;
}

.card p {
    padding: 0 15px;
    margin-bottom: 8px;
}

.facilities {
```

```
    max-width: 600px;
    margin: auto;
    background-color: white;
    padding: 25px 45px;
    border-radius: 8px;
}

.facilities li {
    padding: 7px;
}

.contact {
    background-color: white;
    padding: 25px;
    border-radius: 8px;
    max-width: 650px;
    margin: auto;
}

form {
    max-width: 600px;
    margin: auto;
    background-color: white;
    padding: 25px;
    border-radius: 8px;
}

form label {
    display: block;
    margin-top: 12px;
    font-weight: bold;
}

form input,
form select,
form textarea {
    width: 100%;
    padding: 12px;
    margin-top: 5px;
    border: 1px solid #ccc;
    border-radius: 5px;
}

form textarea {
    height: 100px;
}

form button {
    background-color: #123b70;
    color: white;
    padding: 12px 25px;
    border: none;
    border-radius: 5px;
    margin-top: 15px;
    cursor: pointer;
```

```
}

form button:hover {
    background-color: #2165a8;
}

footer {
    background-color: #08264b;
    color: white;
    text-align: center;
    padding: 20px;
    margin-top: 20px;
}

@media (max-width: 600px) {
    header h1 {
        font-size: 25px;
    }

    nav a {
        display: inline-block;
        margin: 5px 8px;
    }

    .banner {
        padding: 65px 15px;
    }

    .banner h2 {
        font-size: 27px;
    }

    .card {
        width: 100%;
    }

    .container {
        padding: 25px 15px;
    }
}
```

8. OUTPUT SCREENSHOTS & VISUAL LAYOUT REPRESENTATION

The following representations illustrate the browser rendering, structural layout, and user-facing views of the developed pages across desktop and responsive viewports:

Figure: Browser View Layout — Home Page (index.html)
Description: Displays university brand identity, interactive hero banner, academic departments, and campus life overview.

[1] Header & Navigation: Deep Navy bar (#123b70) with 'NRI UNIVERSITY' logo, gold-hover navigation links:

Home | Courses | Admissions.
[2] Hero Banner: Darkened campus background photography with headline 'Welcome to NRI University' and gold CTA button 'Apply Now' linking to admissions.html.
[3] About Section: Centered institutional vision and educational philosophy text.
[4] Departments Grid: Three-column flexbox card cluster featuring Computer Science, Electronics Engineering, and Mechanical Engineering with imagery and summaries.
[5] Campus Life Grid: Three card components highlighting the Modern Campus, Library facilities, and Student Activities.
[6] Footer: Dark slate (#08264b) bottom banner with copyright year and institution identification.

Figure: Browser View Layout — Courses Page (courses.html)

Description: Presents academic programs with duration metrics, faculty specializations, and campus infrastructure.

[1] Header & Navigation: Synchronized header ('Courses and Faculty') maintaining uniform site-wide navigation.
[2] Academic Programs Grid: Cards showcasing B.Tech CSE, CSE - AI & ML, and Electrical Engineering, each highlighting 4-Year durations.
[3] Faculty Directory: Professional faculty cards for Computer Science Faculty, Engineering Faculty, and Research Faculty.
[4] Campus Facilities Card: Elevated white card housing a styled bulleted list of 5 key facilities (Library, Computer Labs, Sports, Seminar Halls, Activity Spaces).
[5] Footer: Unified footer consistent across all subpages.

Figure: Browser View Layout — Admissions Page (admissions.html)

Description: Guides prospective applicants through the 4-step admissions pipeline and includes an interactive application enquiry form.

[1] Header & Banner: Header ('Your Journey Starts Here') and Admissions 2026-27 hero banner.
[2] Procedure Cards: Sequential 4-card workflow (Step 1: Choose Course, Step 2: Check Eligibility, Step 3: Submit Application, Step 4: Complete Admission).
[3] Upcoming Events: Cards for Technical Seminar, College Fest, and Student Workshop with event photography.
[4] Official Contact Card: Dedicated card containing university contact coordinates (College, Address, Email, Phone).
[5] Admission Enquiry Form: Clean styled input form with fields for Student Name, Email Address, Dropdown Program Selector, Message Textarea, and Submit Enquiry button.
[6] Footer: Consistent institution copyright footer.

9. CONCLUSION

The NRI University Web Development Project successfully accomplishes the objective of establishing a modern, responsive, and user-centric academic portal. By leveraging semantic HTML5 tags and modular CSS3 styling techniques, the application achieves a clean visual aesthetic, high readability, and smooth user flow across the Home, Courses, and Admissions sections.

Key technical takeaways and milestones completed in this project include:

- **Semantic Markup Rigor:** Clean document outlining through semantic elements (<header>, <nav>, <section>, <article>, <footer>) facilitating web accessibility and SEO friendliness.
- **Flexible & Fluid Styling:** Utilization of Flexbox architecture for effortless card alignment, eliminating legacy float hacks and simplifying layout refactoring.
- **Adaptive Multi-Device Experience:** Smooth responsive scaling across smartphones, tablets, and full desktop displays via optimized CSS media queries.
- **Interactive Data Capture:** Implementation of accessible form controls for prospective student communications.
- **Future Enhancements:** Future iterations can introduce dynamic JavaScript validation, backend database connectivity (Node.js/Python), dynamic course filters, and a real-time student portal.